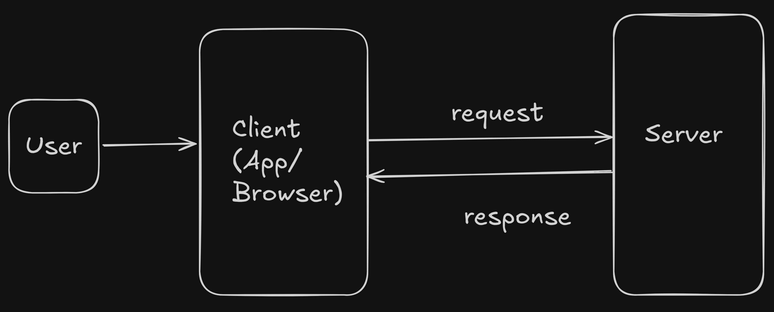
Basic Client Server Architechture

WHat happens when you type abc.com in your browser?



IP address (short for Internet Protocol address)

Types of IP Addresses

1. IPv4 (Internet Protocol version 4)

Format: Four numbers separated by dots (e.g., 192.168.1.1)

Each number (octet): Ranges from 0 to 255

Bit size: 32-bit

Total possible addresses: ~4.3 billion

```
192.168.1.10
↑   ↑   ↑   ↑
A   B   C   D   ← 4 octets
```

2. IPv6 (Internet Protocol version 6)

Format: Eight groups of hexadecimal numbers separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)

Bit size: 128-bit

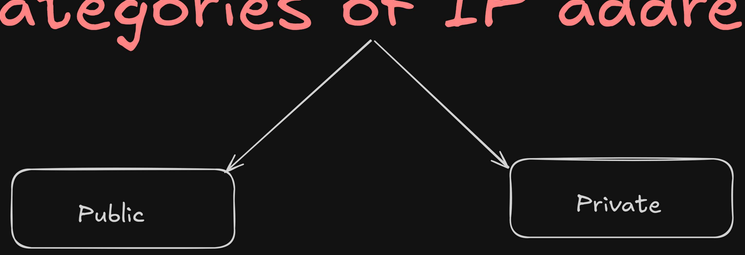
Total addresses: ~340 undecillion (massively more than IPv4)

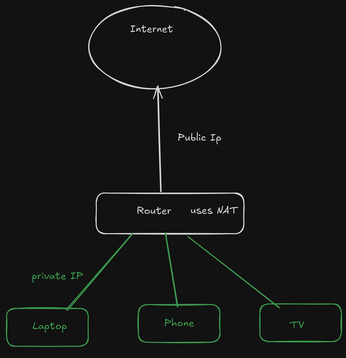
Why IPv6?

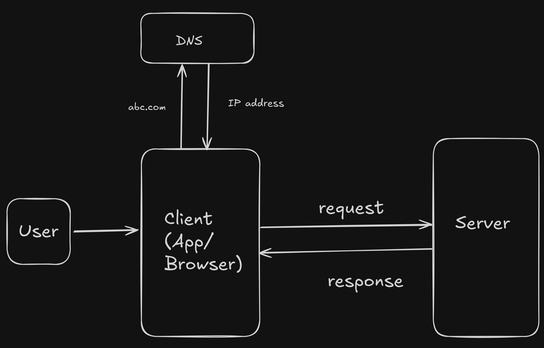
IPv4 addresses are running out due to the growing number of devices.

IPv6 solves this by offering a much larger address space.

Categories of IP address





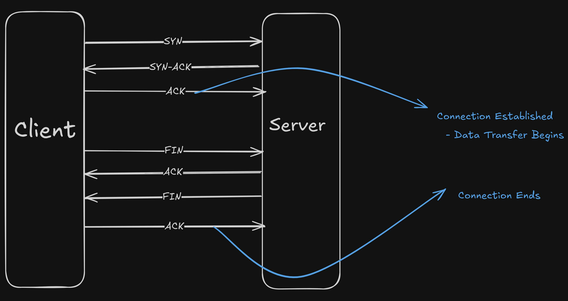


OSI Model - 7 Layers

Transport Layer → TCP/UDP

Application Layer → HTTP/HTTPS, Websockets, WebRTC

TCP



Key Features of TCP

Connection-oriented: Establishes a connection before data transfer.

Reliable: Guarantees delivery by retransmitting lost packets.

Ordered: Data packets arrive in the correct sequence.

Error checking: Detects errors and corrects them.

Flow control: Manages data rate to avoid overwhelming the receiver.

Congestion control: Adjusts the sending rate to avoid network congestion.

UDP - User Datagram Protocol.

Type: Connectionless

No Ordering, some dataloss acceptable

Purpose: Fast transmission without guaranteed delivery

Use case: When speed is more important than reliability

Example: Live streaming, online gaming